

Instruction for MAPP-ASR Kazakh

MAPP ASR — Post-processing and Evaluation of Kazakh ASR Transcription
(based on Google Colab + CSE morphology)

1. Setting Up the Google Colab Environment

1. Open a new notebook in Google Colab.
2. Go to Runtime → Change runtime type.
3. Select Hardware accelerator: GPU (recommended to speed up sentence-transformers).
4. Click Save.

2. Mounting Google Drive and Setting the Working Directory

Run the following cell to mount Google Drive and navigate to the project directory:

```
from google.colab import drive
drive.mount('/content/drive')

%cd /content/drive/MyDrive/Colab\ Notebooks/dataset_wav16k
```

Note: Make sure the dataset_wav16k directory contains all required input files (see Section 4).

3. Installing Required Libraries

In a separate cell, install the following packages:

```
!pip install -q openai sentence-transformers openpyxl xlrd==2.0.1
!pip install -q openai-whisper jiwer sacrebleu bert-score
!apt-get -y -qq install ffmpeg
```

The remaining libraries (re, os, numpy, math, dataclasses, typing, etc.) are part of the standard Python/Colab environment.

4. Preparing Input Files

Ensure the following files are present in the working directory:

File name	Description
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qaz_stems_unik_edit.xlsx	List of Kazakh stems in .xlsx format
qaz_endings_morph_5994.xls	Endings and their morphological descriptions (.xls format)
stop_words.txt	Stop-word list (one word per line, UTF-8)
1m/1m_001.wav	Input audio file for transcription (WAV, 16 kHz)

Note: Audio files must be located in the 1m/ subdirectory.

5. Configuring Paths and Parameters

Verify the following variables at the top of the script:

```

BASE_DIR          = "/content/"
STEMS_XLSX        = "/content/drive/MyDrive/Colab
Notebooks/dataset_wav16k/qaz_stems_unik_edit.xlsx"
ENDINGS_XLS       = "/content/drive/MyDrive/Colab
Notebooks/dataset_wav16k/qaz_endings_morph_5994.xls"
STOPWORDS_TXT     = "/content/drive/MyDrive/Colab
Notebooks/dataset_wav16k/stop_words.txt"
AUDIO_MP3         = "/content/drive/MyDrive/Colab
Notebooks/dataset_wav16k/1m/1m_001.wav"
USE_EMBEDDINGS    = True    # semantic candidate scoring via sentence-
transformers

```

If sentence-transformers is unavailable, set `USE_EMBEDDINGS = False`. In this case, candidate selection will rely solely on Levenshtein distance and CSE scoring.

6. Running the Program

6.1. Loading Resources

Run cells 3–5 sequentially. The program will load:

- stems from qaz_stems_unik_edit.xlsx,
- endings and morphological descriptions from qaz_endings_morph_5994.xls,
- stop words from stop_words.txt.

Upon completion, the following will be printed:

```

Loaded:
  stems:      <N>
  endings:    <N>

```

```
stopwords: <N>
```

6.2. Loading the Whisper Model

Local audio transcription uses OpenAI Whisper. Select the model size:

```
whisper_model = whisper.load_model("medium")    # recommended
# whisper_model = whisper.load_model("large")    # for highest quality
```

6.3. Audio Transcription

Run the transcription cell:

```
raw_text = transcribe_whisper_local(AUDIO_MP3)
print("RAW ASR:", raw_text)
```

6.4. Post-processing (MAPP)

Apply the MAPP pipeline to correct ASR errors:

```
fixed_text = mapp_asr_process(raw_text)
print("FIXED:", fixed_text)
```

The `mapp_asr_process` function performs:

- tokenization with punctuation preservation,
- stem + ending lookup using the CSE model,
- phonetic variants for ASR errors (e.g. substitution of *ʏ*/*y*, *ɪ*/*i*, etc.),
- correction via Levenshtein distance (maximum 1–2 edits),
- semantic selection of the best candidate via sentence-transformers (if `USE_EMBEDDINGS = True`),
- final assembly with punctuation rules.

7. Computing Quality Metrics

After post-processing, compute metrics comparing the output against a reference text. The program calculates the following:

Metric	Description
WER	Word Error Rate – proportion of incorrect words
CER	Character Error Rate – proportion of incorrect characters
BLEU	Bilingual Evaluation Understudy (n-gram precision)

chrF	Character n-gram F-score
TER	Translation Edit Rate
BERTScore	Semantic similarity via BERT vectors (language: kk)

Note: All metrics are computed for both RAW (original ASR output) and FIXED (after MAPP), allowing the contribution of post-processing to be measured.

8. Output Structure

The program prints results to the console in the following format:

```
REF   : <reference text>
RAW   : <raw ASR text>
FIXED : <text after MAPP>
```

9. Notes on Reproducibility

- The CSE morphological logic is deterministic and requires no training.
- Results with USE_EMBEDDINGS = True may vary slightly depending on the version of sentence-transformers.
- The phrase replacement list (PHRASE_REPLACEMENTS) and correction dictionary (WORD_MAP) can be extended for a specific corpus.

10. License and Usage

- Software: MIT License.
- Linguistic resources (stem and ending dictionaries): CC BY 4.0.
- Intended for research and reproducibility purposes.